

Customer Segmentation & Churn Pattern Analytics in European Banking

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Abstract

Customer churn remains one of the most pressing challenges in the banking sector. This study explores churn patterns using a European bank dataset, applying segmentation techniques and visual analytics to uncover demographic, financial, and behavioral drivers of customer attrition. The findings highlight key risk groups and provide actionable insights for customer retention strategies. This project was carried out under the guidance of Mr. Saiprasad Kagne.

Introduction

Banks face significant losses when customers discontinue their services. Understanding who is leaving and why is critical for designing effective retention programs. This project uses a publicly available dataset to analyze churn behavior, focusing on demographic, financial, and engagement variables. The goal is to present findings in a way that balances technical rigor with practical readability. The mentorship of Mr. Saiprasad Kagne provided valuable direction in structuring the analysis and ensuring its relevance to real-world banking challenges.

Methodology

The analysis was conducted using Python with libraries such as pandas, numpy, matplotlib, and seaborn. The workflow included:

- Data Cleaning: Removal of irrelevant fields (e.g., surnames), handling missing values, and ensuring binary consistency in categorical variables.
- Segmentation: Creation of derived fields such as Age Groups, Credit Score Bands, Tenure Groups, and Balance Segments.
- Exploratory Analysis: Calculation of churn rates across segments, visualization of churn distribution, and comparative profiling of retained vs. churned customers.

Results

Overall Churn

- The overall churn rate was approximately 20%, meaning one in five customers exited.
- A pie chart visualization confirmed the imbalance between retained and churned customers.

Segment-Level Insights

- Age Groups: Customers aged 30–45 showed higher churn rates compared to younger and older groups.

- Geography: Certain regions exhibited disproportionately high churn, suggesting localized risk factors.
- Balance Segments: Customers with zero balances were more likely to churn, while surprisingly, even high-balance customers showed notable attrition.
- Tenure Groups: Mid-term customers (3–7 years) churned more than new or long-term customers.

Contribution Analysis

- Age and Credit Score contributed significantly to churn volume.
- High-value customers (large balances) accounted for a substantial share of lost revenue, underscoring the financial risk of churn.

Comparative Profiles

- Churned customers tended to have lower credit scores, fewer products, and lower engagement levels.
- Gender differences were modest but present, with one gender showing slightly higher churn risk.
- Inactive members were far more likely to churn compared to active ones.

My Analysis

The following key takeaways summarize the most critical insights derived from this project:

1. Mid-tenure, middle-aged customers with low balances and low credit scores are the most vulnerable to churn.
2. Inactive customers churn far more than active ones → engagement programs are critical.
3. Zero-balance customers are the largest churn group, but high-balance churners pose the biggest revenue risk.
4. Geography and gender differences matter → churn prevention strategies should be tailored by demographic.
5. Financial stability indicators (credit score, salary, balance) strongly correlate with churn.

Conclusion

This project provides a clear, segmentation-driven understanding of customer churn in European banking. By uncovering churn patterns across geography, demographics, and financial profiles, it equips decision-makers with actionable insights to design targeted, data-driven retention strategies. The successful completion of this project was made possible through the guidance and mentorship of Mr. Saiprasad Kagne.

References

1. Gupta, S., & Lehmann, D. R. (2003). Customers as assets. *Journal of Interactive Marketing*, 17(1), 9–24.
2. Verbeke, W., Martens, D., & Baesens, B. (2011). Social network analysis for customer churn prediction. *Applied Soft Computing*, 11(3), 221–230.
3. European Banking Authority (2024). Customer retention and risk management in EU banks. *Annual Report*.